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THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 15: Case Study

By TAs

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Magnetic Levitation System



- **What is a Magnetic Levitation System?**
 - A system that uses magnetic force to levitate objects in the air.
 - Widely used in high-speed trains, precision instruments, and other fields.
- **System Components:**
 - **Coil:** Generates a magnetic field when electrified.
 - **Steel Ball:** Levitates under the influence of the magnetic force.
 - **Sensor:** Detects the position of the steel ball.
 - **Controller:** Adjusts the coil current to maintain stable levitation of the steel ball.

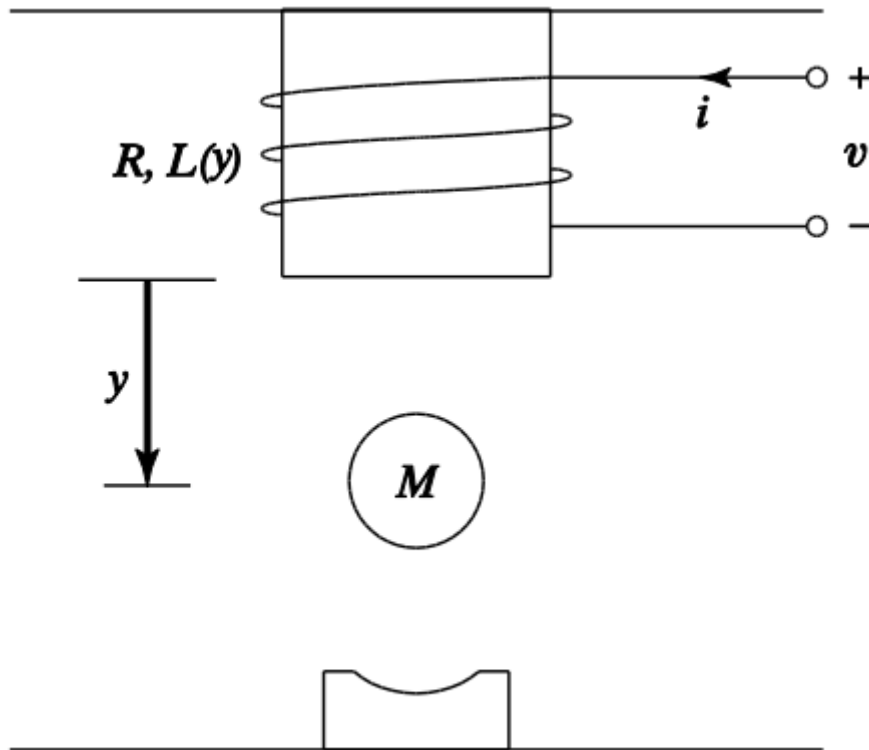
Magnetic Levitation System



- **Task:** suspend the steel ball at **a desired position**.
- **Questions:** SISO, SIMO, MISO or MIMO?
Stabilization or regulation?

SISO

Electromechanical Systems



- The **input** voltage $v(t)$ and **output** position $y(t)$
- **Intermediate variables:** coil current $i(t)$,
ball position $y(t)$
- **Physical laws:** Kirchhoff's voltage law (KVL),
Newton's second law

Electromechanical Systems

- The approximate **inductance** function $L(y)$:

Based on Empirical studies
$$L(y) = L_0 + \frac{rL_1}{y}$$

where r is the radius of the ball, and L_0 and L_1 are constants.

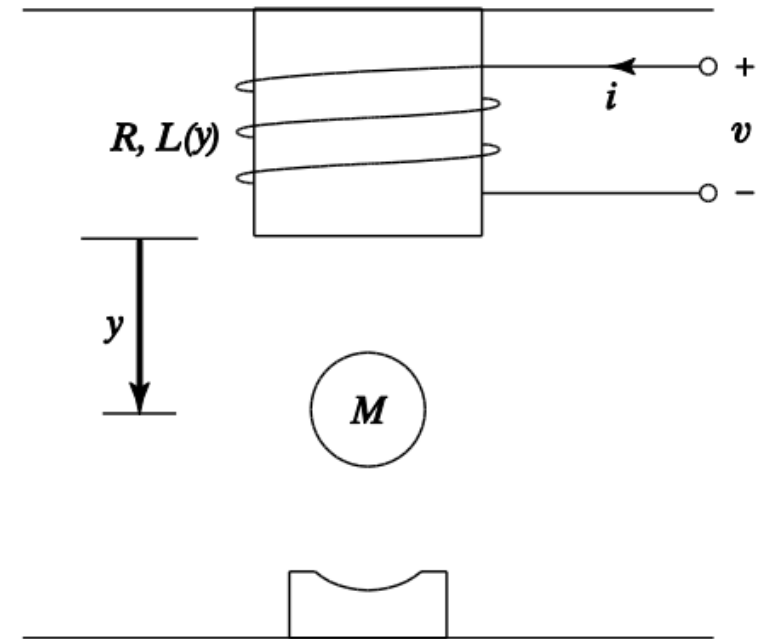
- KVL applied to the coil:

$$Ri(t) + \frac{dL(y(t))i(t)}{dt} = v(t).$$

- Break the derivative of the product:

$$Ri(t) - \frac{rL_1 i(t)}{y^2(t)} \frac{dy(t)}{dt} + L(y(t)) \frac{di(t)}{dt} = v(t).$$

$$v = \frac{d\psi}{dt} + iR$$



Modeling of the lifting force

The total electrical power used to sustain the magnetic field and to move the steel ball is

$$i(t) \frac{dL(y(t))i(t)}{dt} = i^2(t) \frac{dL(y(t))}{dt} + i(t)L(y(t)) \frac{di(t)}{dt}.$$

The rate of change of the magnetic energy stored in the magnetic field is

$$\frac{d}{dt} \left[\frac{1}{2} L(y(t)) i^2(t) \right] = \frac{1}{2} i^2(t) \frac{dL(y(t))}{dt} + L(y(t)) i(t) \frac{di(t)}{dt}$$

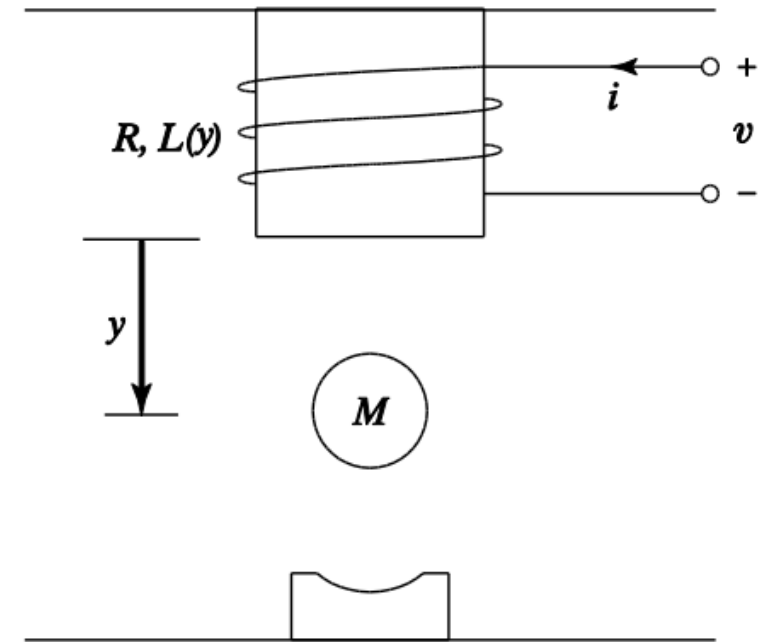
and the mechanical power output is $-f(t) \frac{dy(t)}{dt}$. The minus sign here is due to the opposite directions chosen for $f(t)$ and $y(t)$. Due to the conservation of energy, we have

$$i^2(t) \frac{dL(y(t))}{dt} + i(t)L(y(t)) \frac{di(t)}{dt} = \frac{1}{2} i^2(t) \frac{dL(y(t))}{dt} + L(y(t)) i(t) \frac{di(t)}{dt} - f(t) \frac{dy(t)}{dt}.$$

After canceling the equal terms in both sides, we obtain

$$f(t) = - \frac{1}{2} i^2(t) \frac{dL(y(t))}{dy} = \frac{r L_1}{2} \frac{i^2(t)}{y^2(t)}.$$

which relates the lifting force with the coil current and the ball position.



The electrical energy input to the system is equal to the sum of the change in magnetic energy and the mechanical energy output.

Electromechanical Systems

to the top

- The **lifting force** with the coil current i and position y :

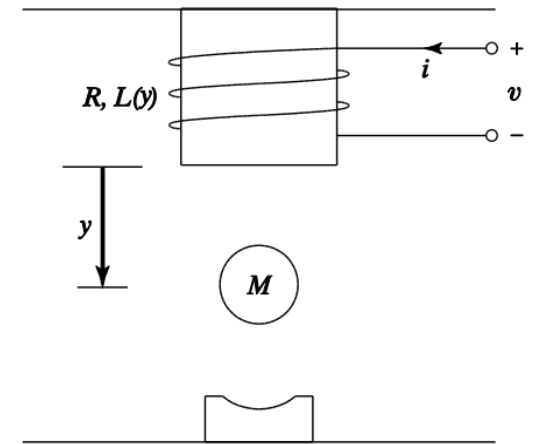
$$f(t) = \frac{rL_1}{2} \frac{i^2(t)}{y^2(t)}.$$

The magnetic force is directly proportional to the square of the coil current and inversely proportional to the square of the ball's position.

- Newton's second law applied to the ball:

$$M \frac{d^2 y(t)}{dt^2} = -\frac{rL_1}{2} \frac{i^2(t)}{y^2(t)} + Mg.$$

- The magnetic levitation system has the differential equation model. The variable $y(t)$ and $i(t)$ are involved nonlinearly in the equations.



Nonlinear system!!

Linearization

- **Why is Linearization Necessary?**

- Nonlinear systems are difficult to analyze and control directly.
- Linearization can approximate the system behavior near the operating point.

- **Operating Point Selection:**

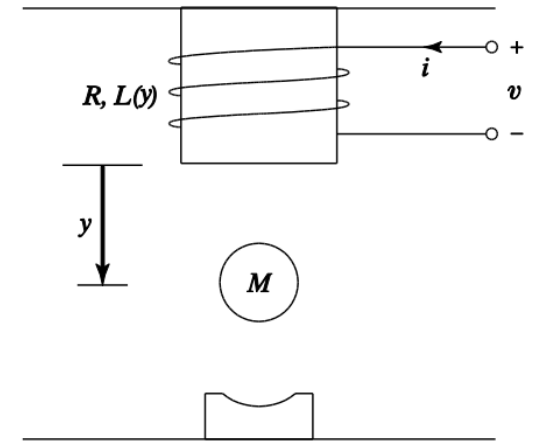
- Choose the equilibrium position y_0 where the steel ball is levitated.
- Perform linearization around this equilibrium position.

State Space Model

- Linearized magnetic levitation system

Choose $x_1(t) = i(t)$, $x_2(t) = y(t)$, $x_3(t) = \dot{y}(t)$ and $u(t) = v(t)$:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} \frac{u(t) - Rx_1(t) + rL_1x_1(t)x_3(t)/x_2^2(t)}{L(x_2(t))} \\ x_3(t) \\ -\frac{rL_1}{2M} \frac{x_1^2(t)}{x_2^2(t)} + g \end{bmatrix}$$
$$y(t) = x_2(t).$$



State space equation

Linearization

- **Operating Point:**

Task: lift the ball to a certain height and suspend it at that height.

Hence we wish to linearize it around an operating point with $y_t = y_0$

Solve equations

$$0 = \frac{u_0 - Rx_{10} + rL_1x_{10}x_{30}/x_{20}^2}{L(x_{20})}$$

$$0 = x_{30}$$

$$0 = -\frac{rL_1}{2M} \frac{x_{10}^2}{x_{20}^2} + g$$

$$y_0 = x_{20}.$$

Apply a constant voltage to the coil

This gives

$$u_0 = Ry_0 \sqrt{\frac{2Mg}{rL_1}}, \quad \begin{bmatrix} x_{10} \\ x_{20} \\ x_{30} \end{bmatrix} = \begin{bmatrix} y_0 \sqrt{\frac{2Mg}{rL_1}} \\ y_0 \\ 0 \end{bmatrix}, \quad y_0 = y_0.$$

Linearization

- **Linearization Process** Denote the deviations of the input, state, and output variables from the operating point by

$$\begin{aligned}\tilde{u}(t) &= u(t) - u_0 = u(t) - Ry_0\sqrt{\frac{2Mg}{rL_1}} \\ \tilde{\mathbf{x}}(t) &= \mathbf{x}(t) - \mathbf{x}_0 = \begin{bmatrix} x_1(t) - y_0\sqrt{\frac{2Mg}{rL_1}} \\ x_2(t) - y_0 \\ x_3(t) \end{bmatrix} \\ \tilde{y}(t) &= y(t) - y_0.\end{aligned}$$

Linearization

- **Linearization Process** The **linearized model** of the deviation variables is

$$\begin{aligned}\dot{\tilde{\mathbf{x}}}(t) &= \mathbf{A}\tilde{\mathbf{x}}(t) + \mathbf{b}\tilde{u}(t) \\ \tilde{y}(t) &= \mathbf{c}\tilde{\mathbf{x}}(t) + d\tilde{u}(t)\end{aligned}$$

where

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\substack{\mathbf{x}=\mathbf{x}_0 \\ u=u_0}} = \begin{bmatrix} -\frac{R}{L(y_0)} & 0 & \frac{\sqrt{2rL_1Mg}}{y_0L(y_0)} \\ 0 & 0 & 1 \\ -\frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} & \frac{2g}{y_0} & 0 \end{bmatrix}, \quad \mathbf{b} = \left. \frac{\partial \mathbf{f}}{\partial u} \right|_{\substack{\mathbf{x}=\mathbf{x}_0 \\ u=u_0}} = \begin{bmatrix} \frac{1}{L(y_0)} \\ 0 \\ 0 \end{bmatrix},$$
$$\mathbf{c} = \left. \frac{\partial g}{\partial \mathbf{x}} \right|_{\substack{\mathbf{x}=\mathbf{x}_0 \\ u=u_0}} = [0 \quad 1 \quad 0], \quad d = \left. \frac{\partial g}{\partial u} \right|_{\substack{\mathbf{x}=\mathbf{x}_0 \\ u=u_0}} = 0.$$

Transfer function

- Using the transfer function formula:

$$P(s) = \mathbf{c}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b} + d$$

$$P(s) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s + \frac{R}{L(y_0)} & 0 & -\frac{\sqrt{2rL_1Mg}}{y_0L(y_0)} \\ 0 & s & -1 \\ \frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} & -\frac{2g}{y(0)} & s \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ \frac{1}{L(y_0)} \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{-\frac{1}{y_0}\sqrt{\frac{2rL_1g}{M}} \frac{1}{L(y_0)}}{\left(s + \frac{R}{L(y_0)}\right) \left(s^2 - \frac{2g}{y_0}\right) + \frac{2rL_1g}{y_0^2L(y_0)}s}.$$

Determines the zeros of the system, which influence the dynamic response.

Determines the poles of the system, which dictate system stability.

Controller Design

Objectives:

- Though linearization, we obtain a third-order system $P(s)$.
- Design a second-order bi-proper controller $C(s)$ to stabilize the system.
- The input to the controller is the deviation input $\tilde{u}(t)$ of the linearized model, and the output is the deviation output $\tilde{y}(t)$.

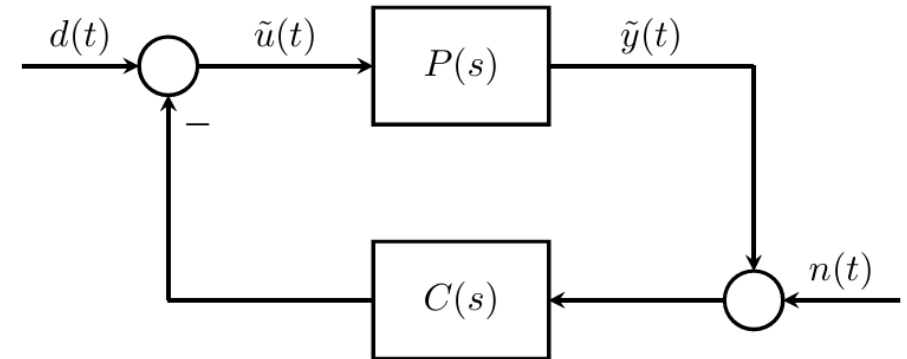
Pole Placement Method:

- The system is third-order, and the controller is second-order, resulting in a total of five closed-loop poles.
- The locations of these five poles determine the overall dynamic response of the system.

Controller system structure

- **Linear Model Control System Structure:**

The control system for the linear model includes the controller $C(s)$, disturbance $d(t)$, and sensor noise $n(t)$.

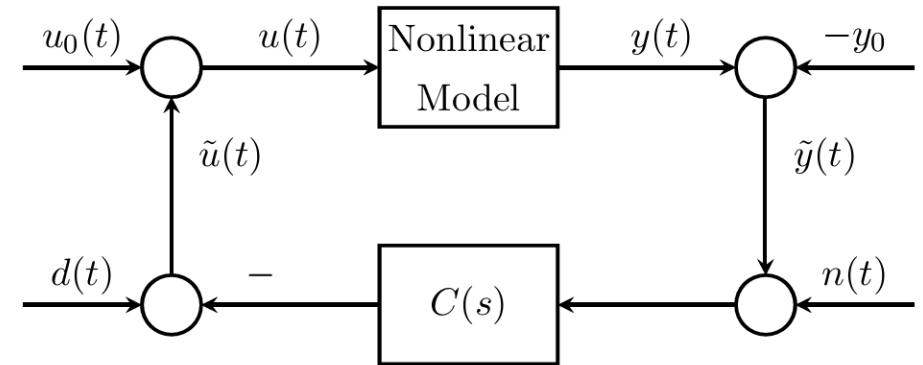


- **Nonlinear Model Control System Structure:**

After we design the controller, we need to apply it in the real nonlinear model to test whether it can stabilize the real system.

Input: $u(t) = u_0 + \tilde{u}(t)$

Output: $y(t) = y_0 + \tilde{y}(t)$



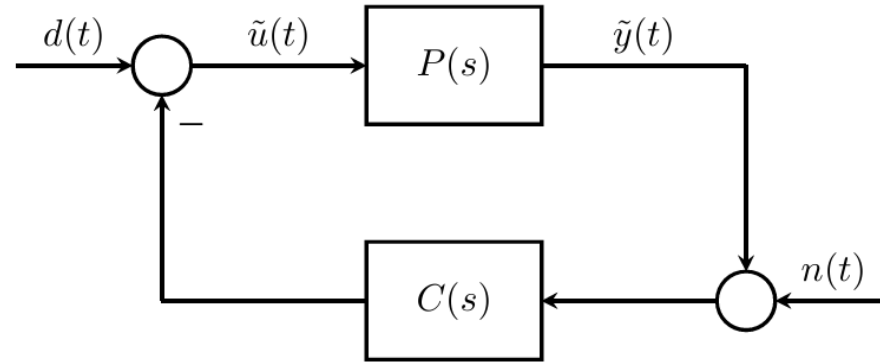
Stabilization of Maglev

➤ **Example** Let $M = 0.1[\text{kg}]$, $R = 10[\Omega]$, $rL_l = 0.002[\text{m}^*\text{H}]$, $y_0 = 0.01[\text{m}]$
 $, L_0 = 0.2[\text{H}]$, $g = 9.8[\text{m}/\text{sec}^2]$

$$C(s) = \frac{-156.5248}{s^3 + 25s^2 - 980s - 49000}$$

- A third order SISO system with poles at 37.0766, $-31.0383 \pm 18.9265i$. **unstable!**

Stabilization of Maglev

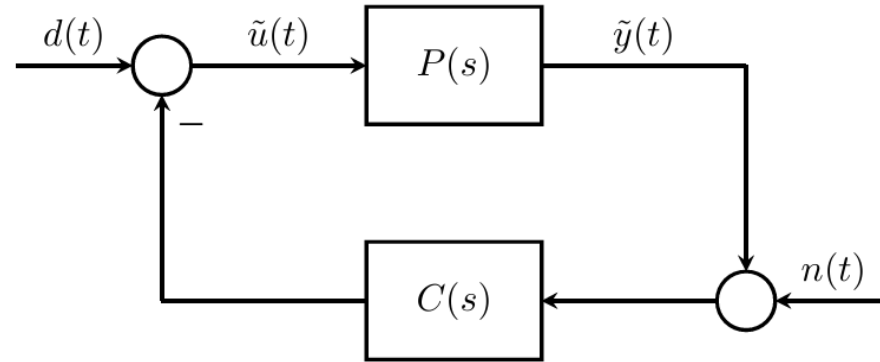


- **Task** Stabilize the system by using the output $\tilde{y}(t)$
- **Pole placement** Design a second-order controller

$$C_x(s) = \frac{q_0 s^2 + q_1 s + q_2}{p_0 s^2 + p_1 s + p_2}$$

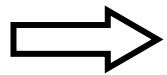
to place all poles to -40 , $-45 \pm 45j$, $-120 \pm 120j$;

Stabilization of Maglev



- This leads to the **Diophantine equation**

$$(s^3 + 25s^2 - 980s + 4900)(p_0s^2 + p_1s + p_2) - 156.5248(q_0s^2 + q_1s + q_2) \\ = (s + 40)(s^2 + 90s + 4050)(s^2 + 240s + 28800)$$



$$\begin{aligned} p_0 &= 1, & p_1 &= 345, & p_2 &= 60005 \\ q_0 &= -29573.44, & q_1 &= -2139660.87, & q_2 &= -48591961.28 \end{aligned}$$

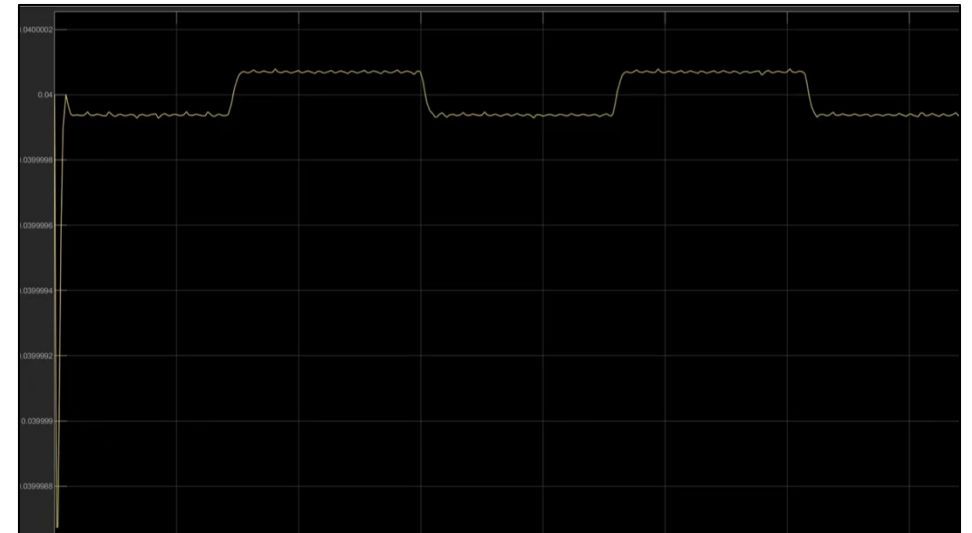
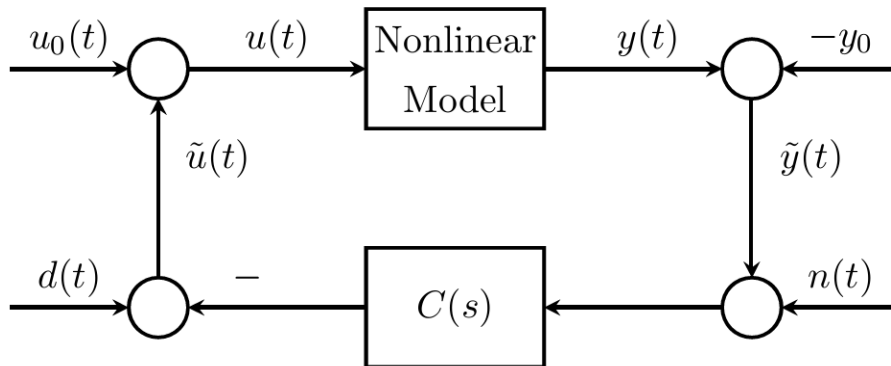
Stabilization of Maglev

- Then the designed controller is

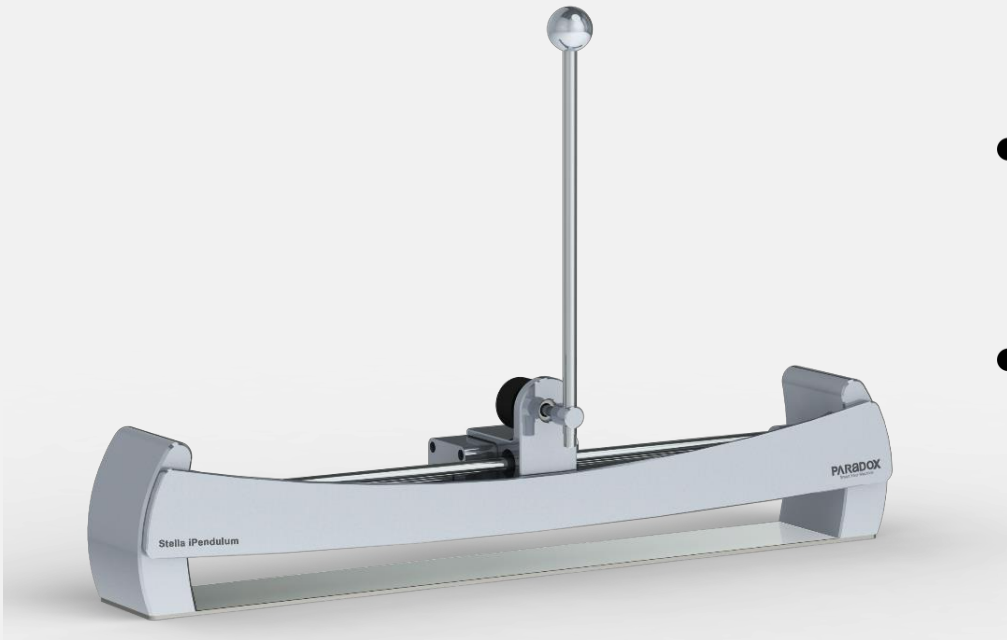
Stable controller!

$$C_x(s) = \frac{-29573s^2 - 2139660s - 48591961}{s^2 + 345s + 60005}$$

- Impulse responses

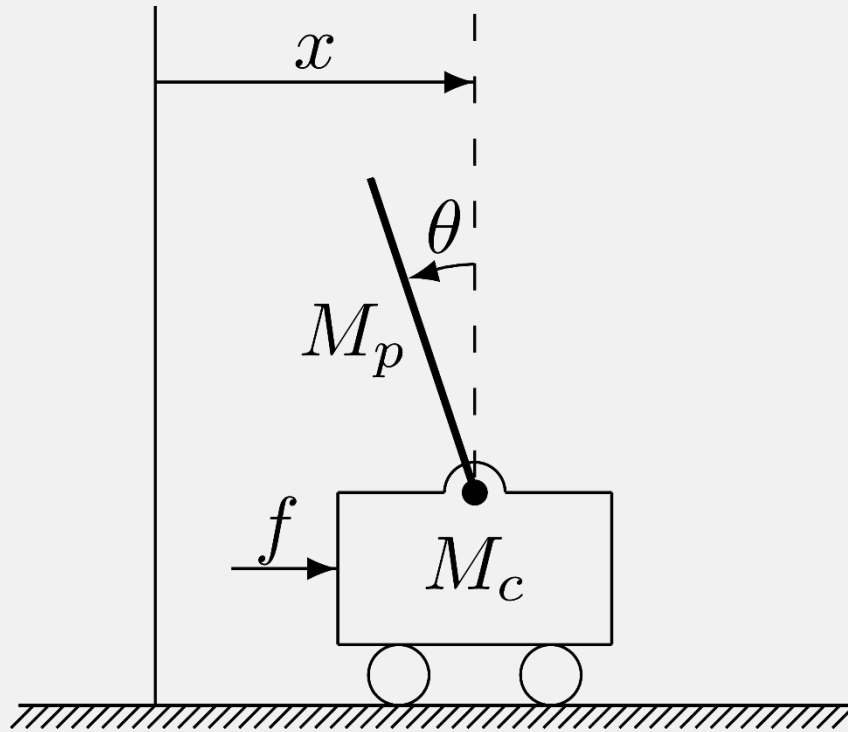


Inverted Pendulum System



- **Task:** keep the pendulum **upward** by moving the car at the base.
- **Mimicry:** balance a long stick **upward** on our finger tip.

Inverted Pendulum System



★Parameters and variables:

M_c : mass of the cart

M_p : evenly distributed mass of the rod

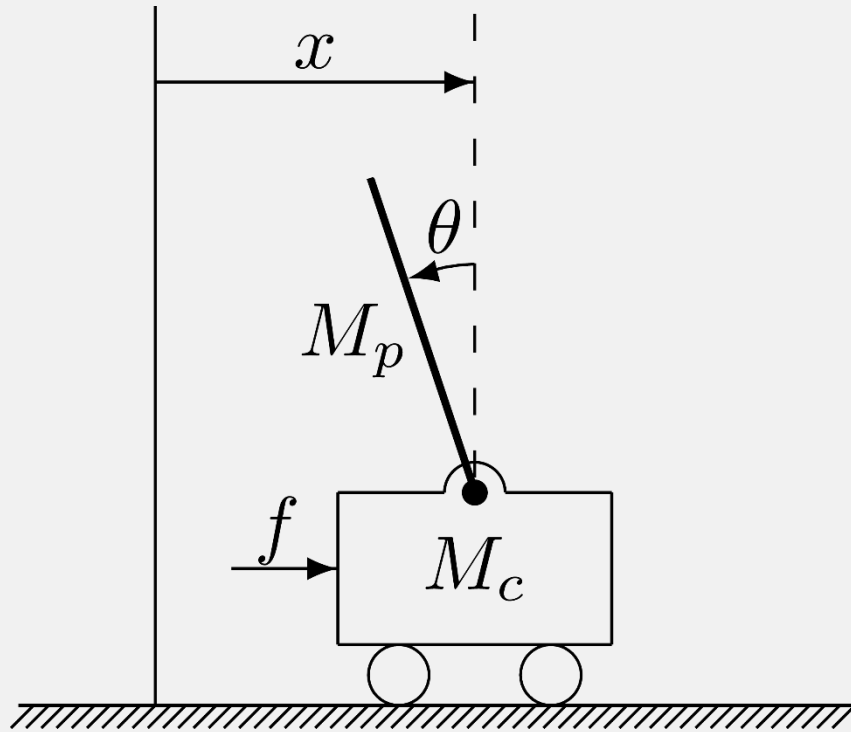
L : length of the rod

$x(t)$: the cart position

$\theta(t)$: the pendulum angle

$f(t)$: the force applied on the cart

Inverted Pendulum System



- The **input** force $f(t)$
- The **two output**
 - the cart position $x(t)$
 - the pendulum angle $\theta(t)$
- **Physical laws:**
 - Newton's second law
 - the rotational version of Newton's second law

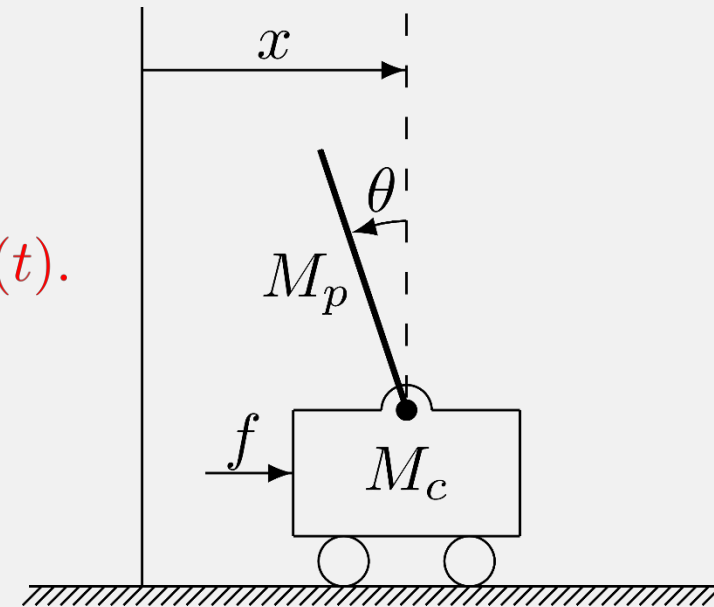
Model of iPendulum

- Apply Newton's law to the **horizontal** direction of the cart

$$\begin{aligned} f(t) &= M_c \frac{d^2 x(t)}{dt^2} + M_p \frac{d^2 [x(t) - L/2 \sin \theta(t)]}{dt^2} \\ &= (M_p + M_c) \ddot{x}(t) - \frac{M_p L}{2} \ddot{\theta}(t) \cos \theta(t) + \frac{M_p L}{2} \dot{\theta}^2(t) \sin \theta(t). \end{aligned}$$

- The **moment of inertia** of the rod w.r.t. the **pivot point** is

$$J = \int_0^L \ell^2 \left(\frac{M_p}{L} d\ell \right) = \frac{1}{3} M_p L^2.$$

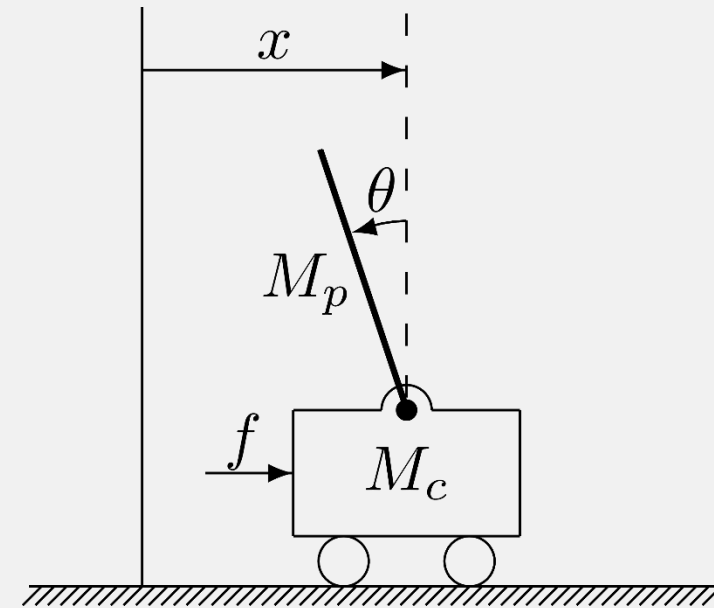


Model of iPendulum

- Apply the rotational version of Newton's second law for the pendulum around the **pivot point**, then

$$J\ddot{\theta}(t) = M_p\ddot{x}(t) \cos \theta(t) \frac{L}{2} + M_p g \sin \theta(t) \frac{L}{2}.$$

$$\Rightarrow \frac{2}{3}L\ddot{\theta}(t) - \ddot{x}(t) \cos \theta(t) - g \sin \theta(t) = 0.$$



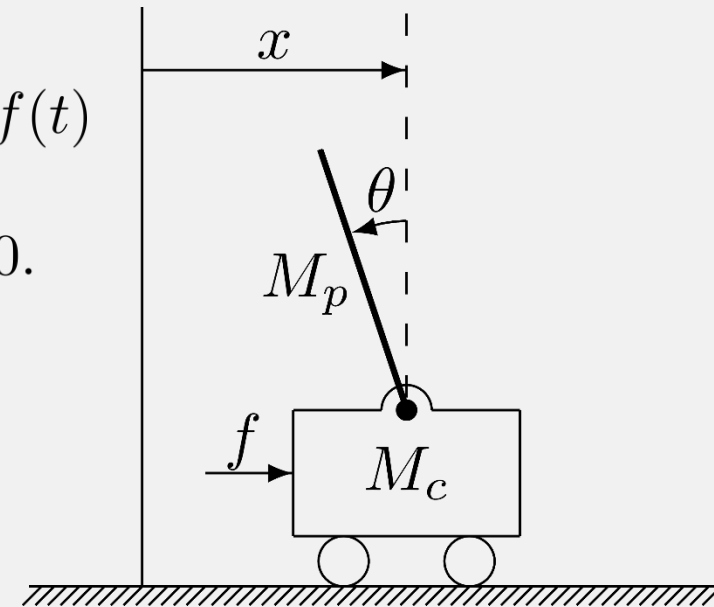
Linearization of iPendulum

- Hence the **differential equation** model of the system:

$$(M_p + M_c)\ddot{x}(t) - \frac{1}{2}M_pL\ddot{\theta}(t)\cos\theta(t) + \frac{1}{2}M_pL\dot{\theta}^2(t)\sin\theta(t) = f(t)$$
$$-\ddot{x}(t)\cos\theta(t) + \frac{2}{3}L\ddot{\theta}(t) - g\sin\theta(t) = 0.$$

- Linearize** it around $x(t) = 0, \dot{x}(t) = 0, \theta(t) = 0, \dot{\theta}(t) = 0$, then

$$(M_p + M_c)\ddot{x}(t) - \frac{1}{2}M_pL\ddot{\theta}(t) = f(t)$$
$$-\ddot{x}(t) + \frac{2}{3}L\ddot{\theta}(t) - g\theta(t) = 0.$$



Nonlinear system!!

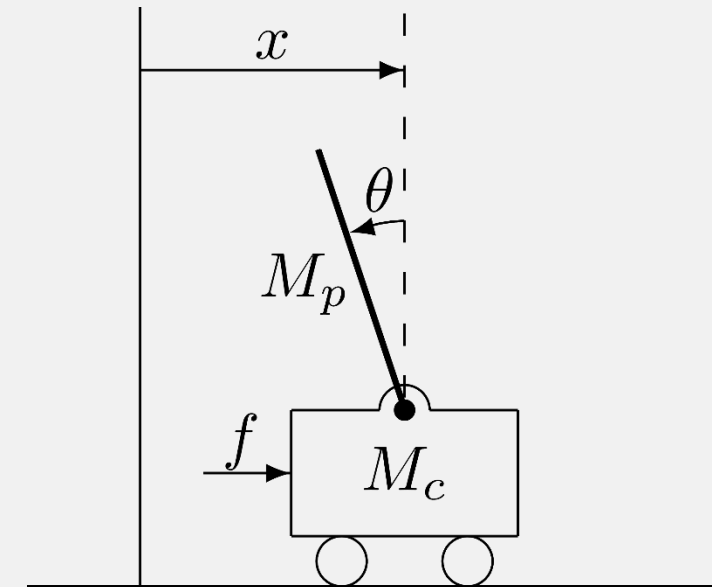
Transfer Function of iPendulum

- Take Laplace transform on both sides of the equations, then

$$(M_p + M_c)s^2 X(s) - \frac{1}{2}M_p L s^2 \Theta(s) = F(s)$$
$$-s^2 X(s) + \frac{2}{3}L s^2 \Theta(s) - g\Theta(s) = 0.$$

- Take $X(s)$ and $\Theta(s)$ as the outputs, and $F(s)$ as the input, then

$$\begin{bmatrix} X(s) \\ \Theta(s) \end{bmatrix} = \frac{1}{(M_p + M_c)s^2(2Ls^2/3 - g) - M_p L s^4/2} \begin{bmatrix} 2Ls^2/3 - g \\ s^2 \end{bmatrix} F(s)$$
$$= \frac{1}{s^2[(M_p + 4M_c)Ls^2 - 6(M_p + M_c)g]} \begin{bmatrix} 4Ls^2 - 6g \\ 6s^2 \end{bmatrix} F(s).$$



Transfer Function of iPendulum

- Hence the vector transfer function of the system is

$$\mathbf{P}(s) = \begin{bmatrix} P_x(s) \\ P_\theta(s) \end{bmatrix} = \frac{1}{s^2[(M_p + 4M_c)Ls^2 - 6(M_p + M_c)g]} \begin{bmatrix} 4Ls^2 - 6g \\ 6s^2 \end{bmatrix}.$$

- **Example** Let $M_c = 1$ [kg], $M_p = 2$ [kg], $L = 1$ [m]. This gives

$$\mathbf{P}(s) = \begin{bmatrix} P_x(s) \\ P_\theta(s) \end{bmatrix} = \frac{1}{s^2(s^2 - 3g)} \begin{bmatrix} 2/3s^2 - g \\ s^2 \end{bmatrix}$$

where $g = 9.8$ [m/sec²].

- A **fourth-order SISO** system with poles at $0, 0, \sqrt{3g}, -\sqrt{3g}$. **Highly unstable!!**

Stabilization of iPendulum

- **Example** The transfer function of iPendulum is

$$P(s) = \begin{bmatrix} P_x(s) \\ P_\theta(s) \end{bmatrix} = \frac{1}{s^2(s^2 - 3g)} \begin{bmatrix} 2/3s^2 - g \\ s^2 \end{bmatrix}$$

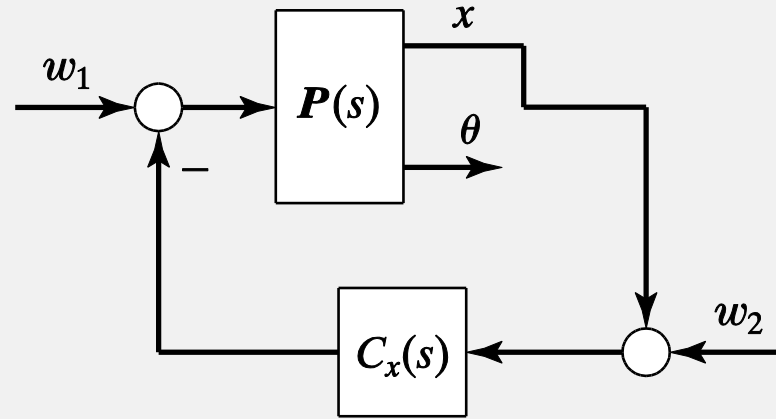
- **Cannot** be stabilized by using **only the $\theta(t)$** output since

$$P_\theta(s) = \frac{1}{s^2 - 3g} = \frac{1}{(s + \sqrt{3g})(s - \sqrt{3g})}. \quad \text{Only one unstable pole.}$$

- **Can** be stabilized by using **only the $x(t)$** output since

$$P_x(s) = \frac{2/3s^2 - g}{s^2(s^2 - 3g)}. \quad \text{Contain all unstable poles.}$$

Stabilization of iPendulum

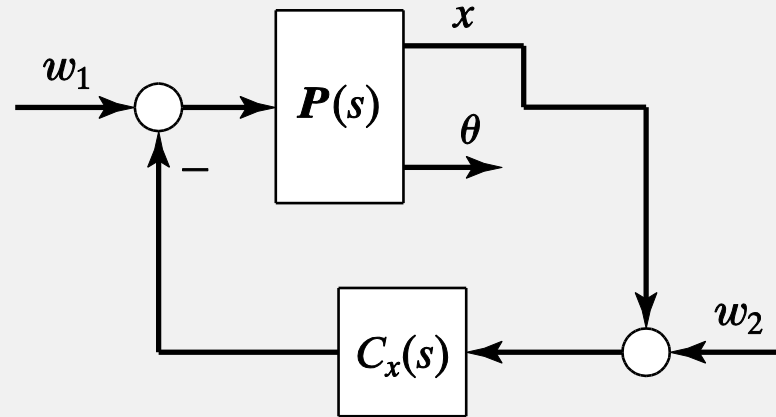


- **Task** Stabilize the iPendulum by only using the output $x(t)$.
- **Pole placement** Design a third-order controller

$$C_x(s) = \frac{q_0 s^3 + q_1 s^2 + q_2 s + q_3}{p_0 s^3 + p_1 s^2 + p_2 s + p_3}$$

to place all poles to $-6, -0.5 \pm j4, -2 \pm j2.5, -8.5 \pm j4.5$.

Stabilization of iPendulum



- This leads to the **Diophantine equation**

$$\begin{aligned} s^2(s^2 - 3g)(p_0s^3 + p_1s^2 + p_2s + p_3) + (0.67s^2 - g)(q_0s^3 + q_1s^2 + q_2s + q_3) \\ = (s + 6)(s^2 + s + 16.25)(s^2 + 4s + 10.25)(s^2 + 17s + 92.5). \end{aligned}$$

$$\begin{aligned} \implies p_0 = 1, \quad p_1 = 28, \quad p_2 = -1434, \quad p_3 = -5963, \\ q_0 = 2705, \quad q_1 = 13640, \quad q_2 = -7567, \quad q_3 = -9433. \end{aligned}$$

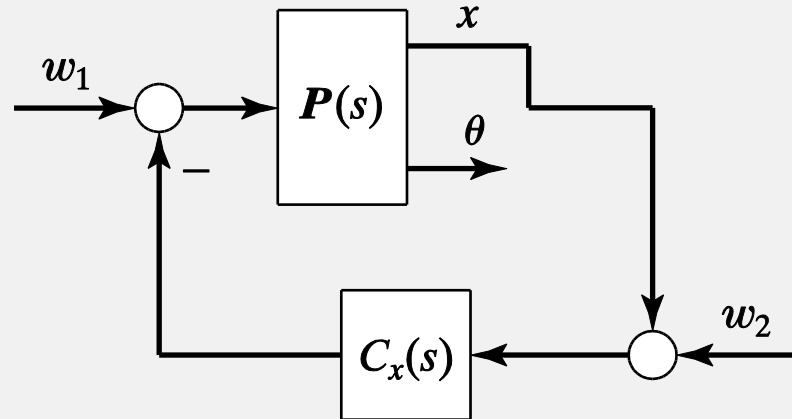
Stabilization of iPendulum

- Then the designed controller is **Extreme values & unstable controller!!**

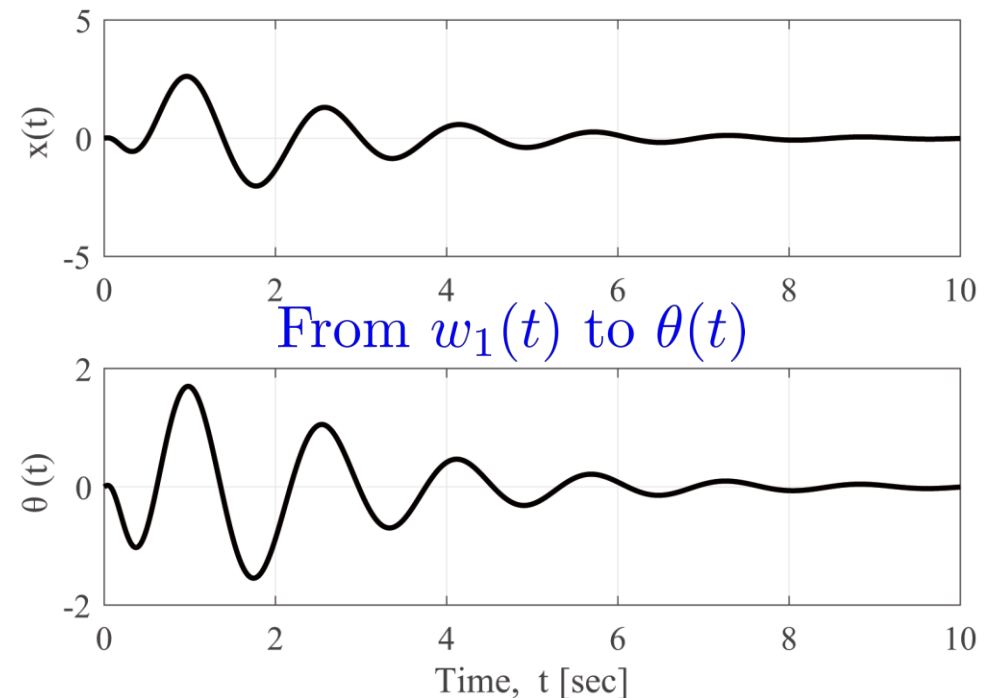
$$C_x(s) = \frac{2705s^3 + 13640s^2 - 7567s - 9433}{s^3 + 28s^2 - 1434s - 5963}.$$

From $w_1(t)$ to $x(t)$

- Impulse responses

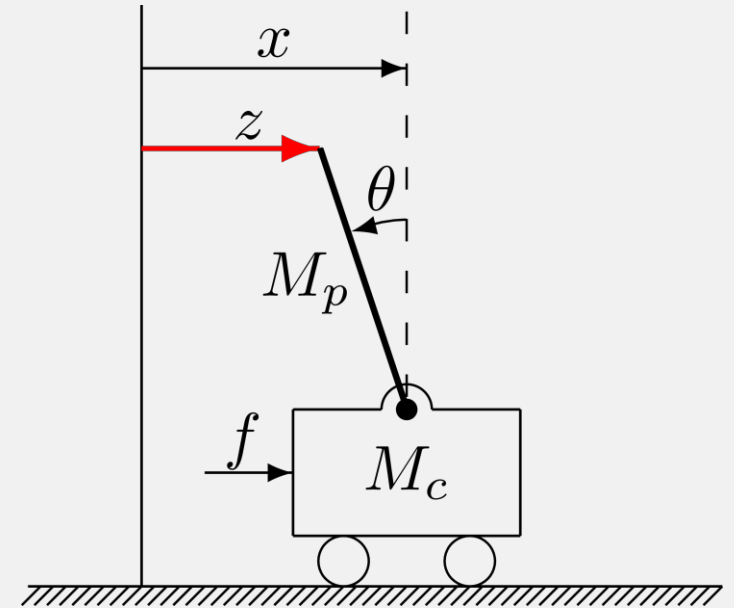
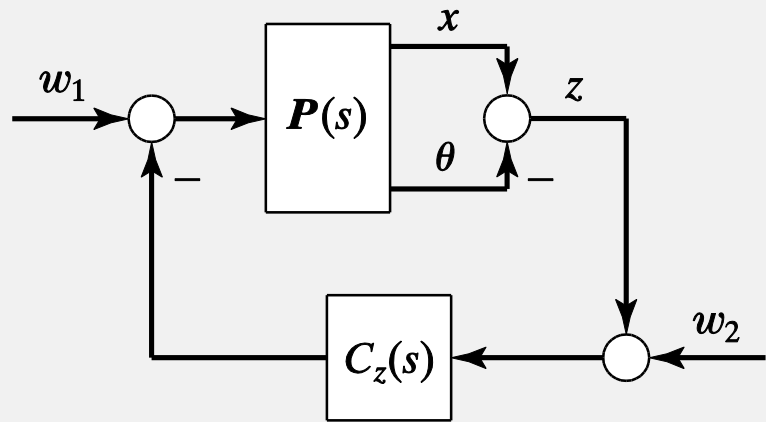


- Not a good design.**



From $w_1(t)$ to $\theta(t)$

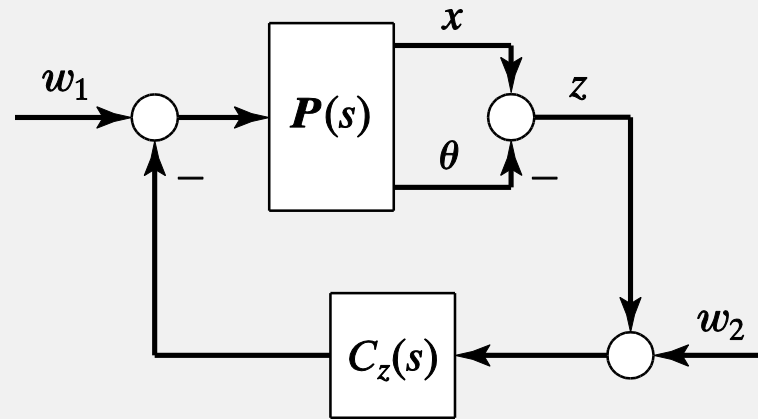
Stabilization of iPendulum



- **Task** Stabilize the iPendulum by using the using output $x(t) - L\theta(t)$.
- **Idea** Look at the **upper tip** of the stick when balance a long stick.
 - Feed back the (horizontal) position of the tip, which is

$$z(t) = x(t) - L \sin \theta(t) \xrightarrow{\text{Linearization}} x(t) - L\theta(t).$$

Stabilization of iPendulum



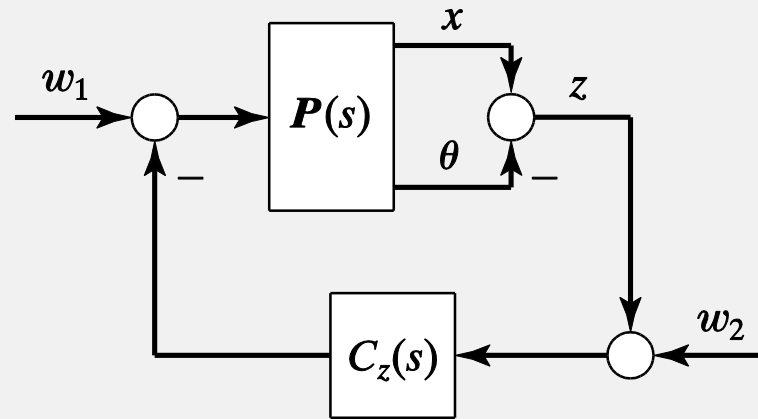
- In this example $L = 1$, take $z(t)$ **as the output**, then the transfer function from $f(t)$ to $z(t)$ becomes

$$P_z(s) = P_x(s) - P_\theta(s) = -\frac{1/3s^2 + g}{s^2(s^2 - 3g)}.$$

- The zero $\sqrt{3g/2}$ of $P_x(s)$ near the pole $\sqrt{3g}$ is no longer a zero of $P_z(s)$.

Much easier to control!!

Stabilization of iPendulum

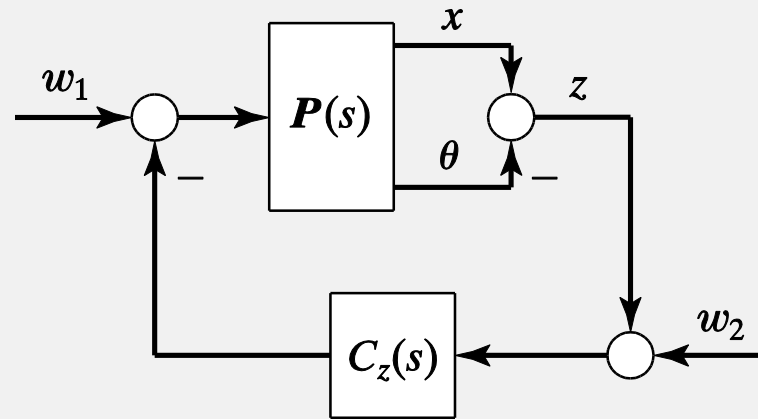


- **Pole placement** Design a third-order controller

$$C_z(s) = \frac{q_0 s^3 + q_1 s^2 + q_2 s + q_3}{p_0 s^3 + p_1 s^2 + p_2 s + p_3}$$

to place the poles to the **same** places $-6, -0.5 \pm j4, -2 \pm j2.5, -8.5 \pm j4.5$.

Stabilization of iPendulum



- The **Diophantine equation** is

$$\begin{aligned} s^2(s^2 - 3g)(p_0s^3 + p_1s^2 + p_2s + p_3) - (1/3s^2 + g)(q_0s^3 + q_1s^2 + q_2s + q_3) \\ = (s + 6)(s^2 + s + 16.25)(s^2 + 4s + 10.25)(s^2 + 17s + 92.5). \end{aligned}$$

$$\begin{aligned} \implies p_0 = 1, \quad p_1 = 28, \quad p_2 = 47.25, \quad p_3 = 1015, \\ q_0 = -966.5, \quad q_1 = -6337, \quad q_2 = -7567, \quad q_3 = -9433. \end{aligned}$$

Stabilization of iPendulum

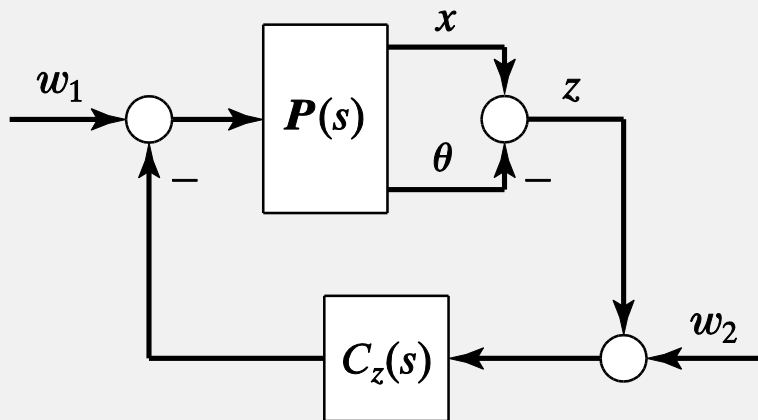
- The designed controller is

Less extreme values & stable controller!!

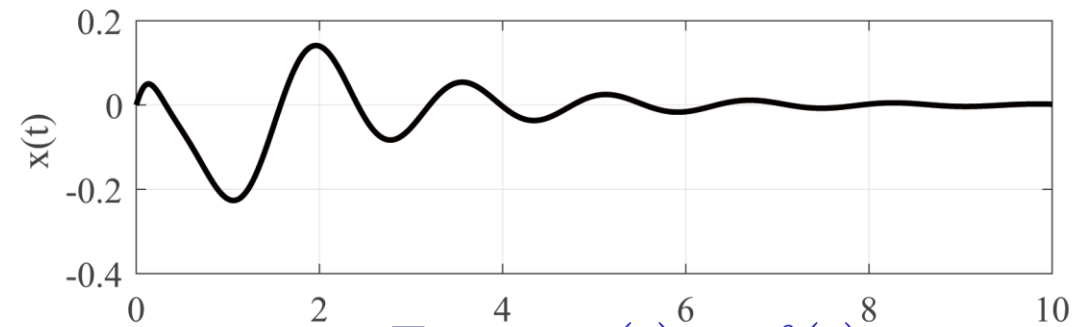
$$C_z(s) = -\frac{966.5s^3 + 6337s^2 + 7567s + 9433}{s^3 + 28s^2 + 47.25s + 1015}.$$

From $w_1(t)$ to $x(t)$

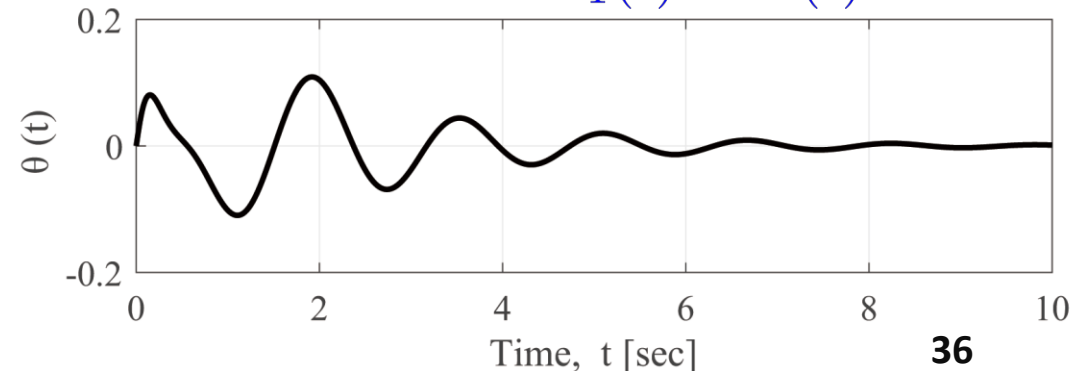
- Impulse responses



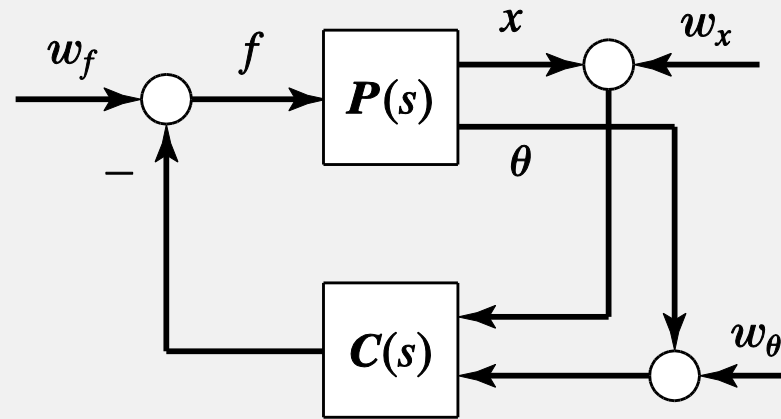
- A better design.**



From $w_1(t)$ to $\theta(t)$

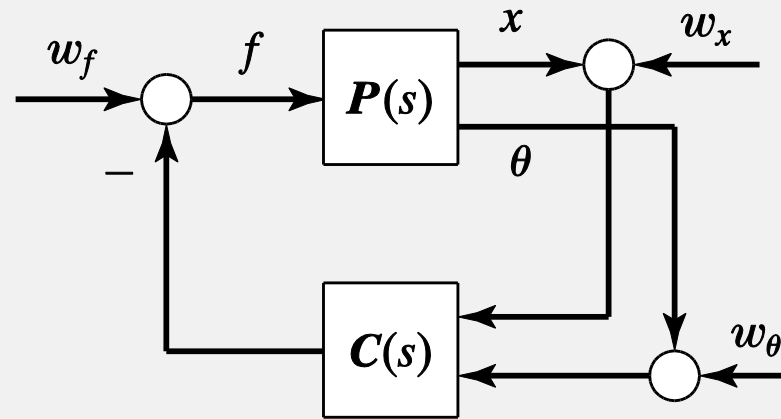


DISO Controller of iPendulum



- The previous two designs are based on **SISO pole placement** theory.
- The iPendulum is a **genuine SIDO** system.
- **Task** Stabilize the system using a **DISO controller** with the smallest order.

DISO Controller of iPendulum



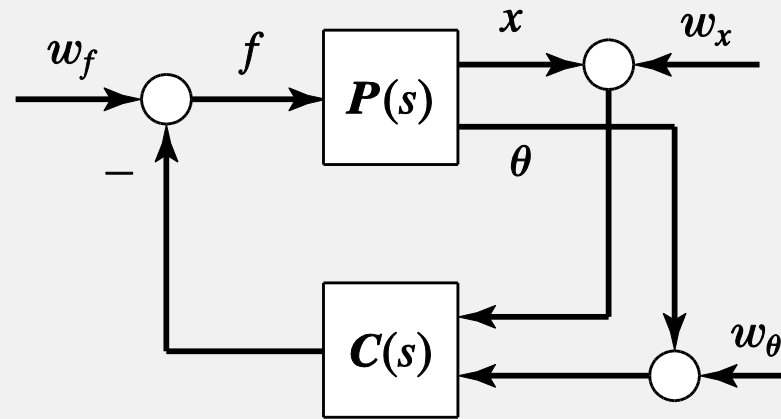
- Let us write the controller as

$$\mathbf{C}(s) = \begin{bmatrix} C_x(s) & C_\theta(s) \end{bmatrix} = \frac{\begin{bmatrix} q_x(s) & q_\theta(s) \end{bmatrix}}{p(s)}.$$

- Then the closed-loop characteristic polynomial is

$$c(s) = a(s)p(s) + b_x(s)q_x(s) + b_\theta(s)q_\theta(s).$$

DISO Controller of iPendulum



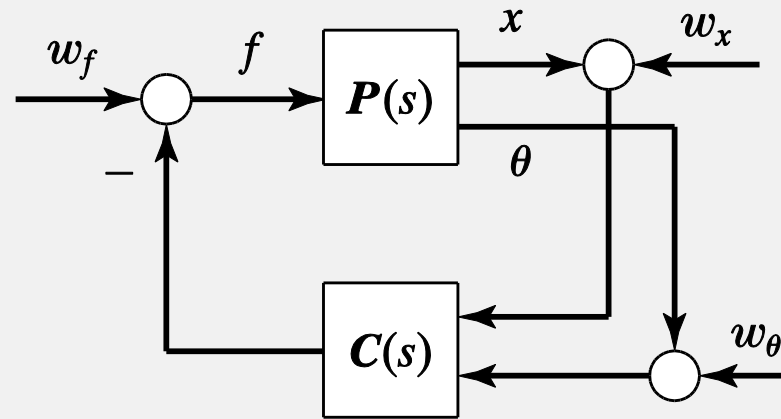
- **Zeroth-order controller** The controller is of the form

$$C(s) = \frac{\begin{bmatrix} q_{x0} & q_{\theta 0} \end{bmatrix}}{p_0}$$

which has **3 parameters**, whereas $c(s) = a(s)p(s) + b_x(s)q_x(s) + b_\theta(s)q_\theta(s)$ has degree 4 with **5 coefficients**.

Cannot work!!

DISO Controller of iPendulum



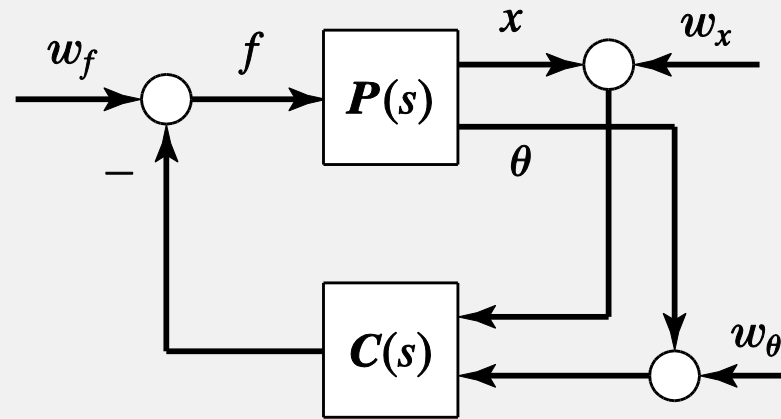
- **First-order controller** The controller is of the form

$$C(s) = \frac{[q_{x0}s + q_{x1} \quad q_{\theta 0}s + q_{\theta 1}]}{p_0s + p_1}$$

which has 6 parameters. **Perfect match!!**

- Place the desired poles to $-6, -2 \pm j2.5, -8.5 \pm j4.5$.

DISO Controller of iPendulum



- **Pole placement** Then the polynomial equation needed is

$$\begin{aligned} s^2(s^2 - 3g)(p_0s + p_1) + (2/3s^2 - g)(q_{x0}s + q_{x1}) + s^2(q_{\theta 0}s + q_{\theta 1}) \\ = (s + 6)(s^2 + 4s + 10.25)(s^2 + 17s + 92.5). \end{aligned}$$

$$\implies p_0 = 1, \quad p_1 = 27, \quad q_{x0} = -430.0, \quad q_{x1} = -580.5, \quad q_{\theta 0} = 612.9, \quad q_{\theta 1} = 2750.$$

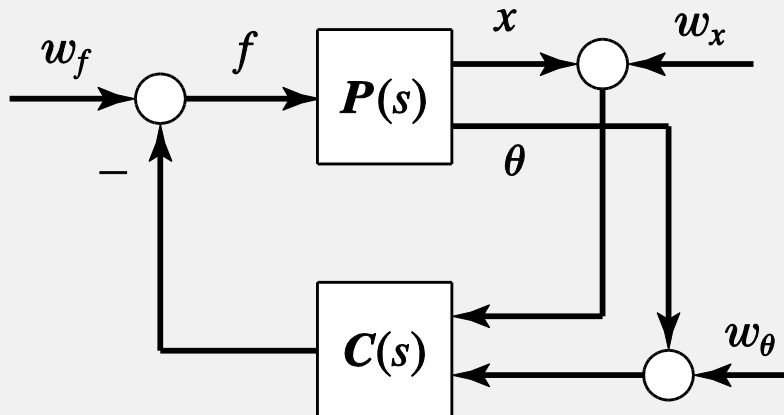
- **Not a Diophantine equation** but can be solved in the same way.

DISO Controller of iPendulum

- Hence the designed controller is **Appropriate values & stable controller!!**

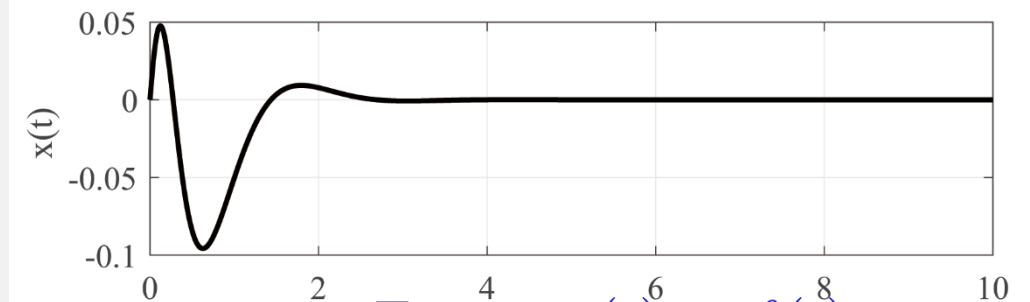
$$C(s) = \frac{[-430.0s - 580.5 \quad 612.9s + 2750]}{s + 27}.$$

- Impulse responses

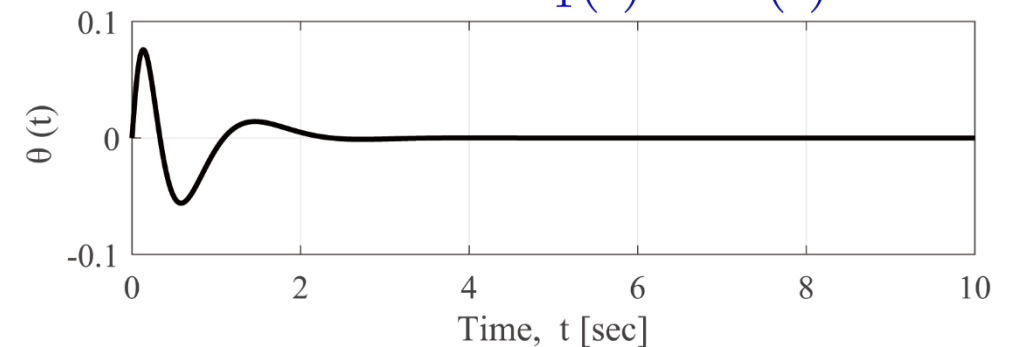


- Better performance.**

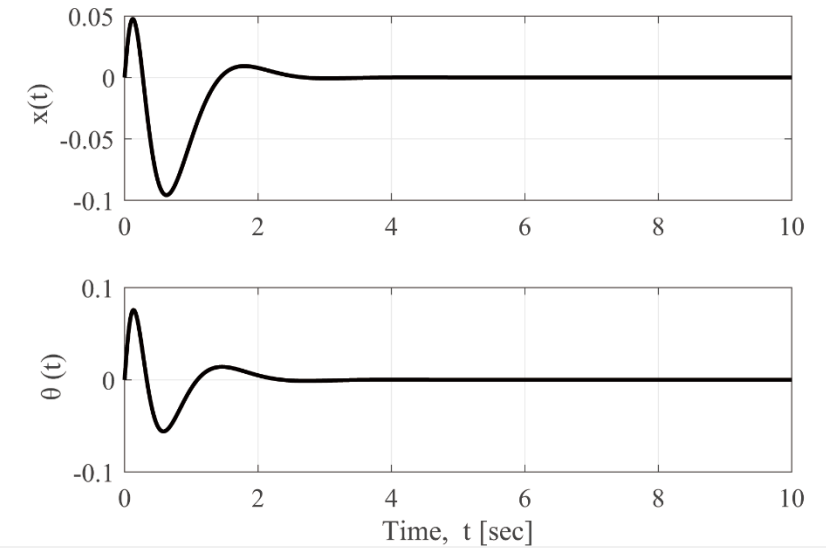
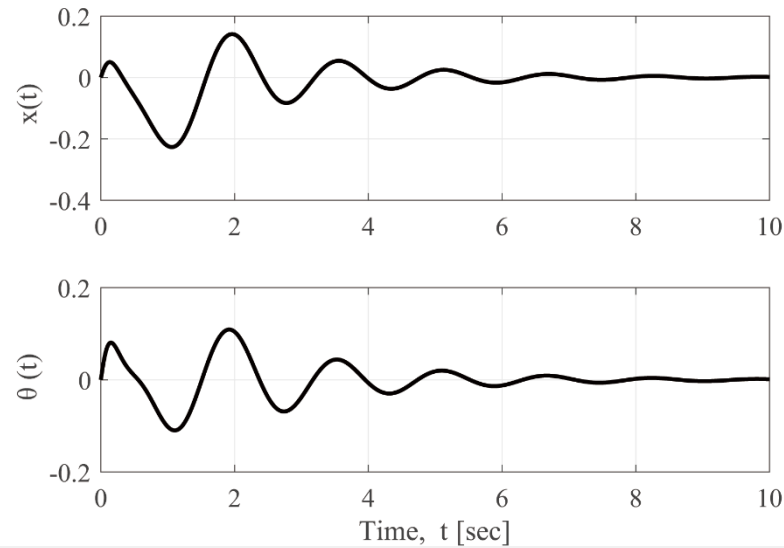
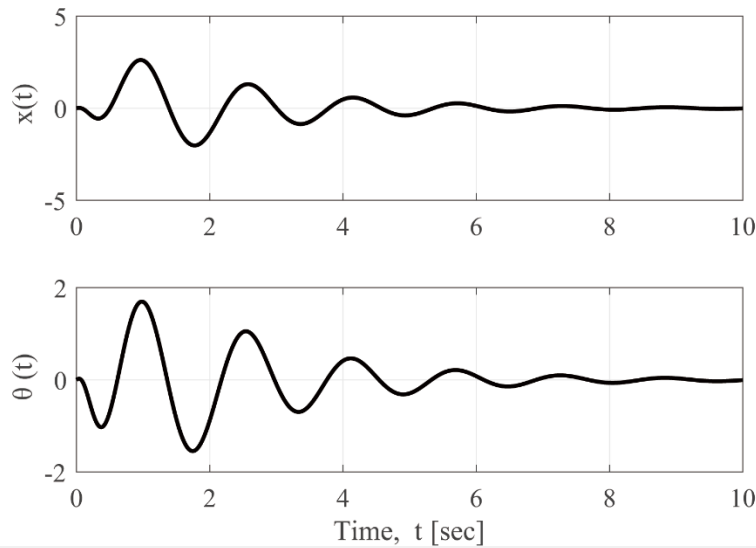
From $w_1(t)$ to $x(t)$



From $w_1(t)$ to $\theta(t)$



Comparison of Three Controllers



- Use only the output $x(t)$
- Use the output $x(t) - L\theta(t)$
- Use the **DISO controller**